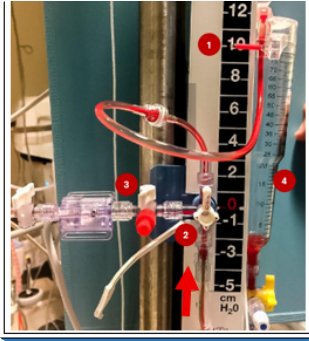


# External Ventricular Drain



## COMPONENTS

1. **Drainage setting:** The ordered height (+5, +10, +20 cmH<sub>2</sub>O) refers to how high the drip chamber is positioned above the external auditory meatus/tragus, the bedside reference for the foramen of Monro. CSF drains only when ventricular pressure exceeds that water-column height, so a lower setting drains more readily than a higher setting.
2. **Drainage stopcock:** 12 o'clock = clamp/closed, 3 o'clock = open to drain.
3. **Transducer and zeroing stopcock:** Controls baseline calibration.
4. **Collection/drip chamber:** Graduated cylinder measuring CSF volume.

\*Red arrow indicates the direction of CSF flow.



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## INDICATIONS

- **CSF Diversion** for acute obstructive hydrocephalus (e.g., IVH, posterior fossa stroke).
- **ICP Monitoring** in severe brain injury (GCS  $\leq$  8).

## SIGNS OF HYDROCEPHALUS

### Clinical Signs:

- Decline in Level of Consciousness (LOC) or progressive somnolence.
- **Parinaud's Syndrome:** Upward gaze palsy (setting sun sign), retraction nystagmus on convergence, and pupillary light-near dissociation.

### Radiographic Signs (NCCT Head):

- Progressive enlargement of the cerebral ventricles.
- Temporal horn dilation (sensitive early sign of obstruction).
- High-risk factors: Intraventricular Hemorrhage (IVH) in 3rd or 4th ventricles, compression of 4th ventricle, or high volume blood.

## BASICS

- **Leveling:** Always align the zero level of the EVD scale/transducer to the external auditory meatus (EAM) / tragus.
- **Mobilization Clamping Rules:** Always CLAMP the EVD before: turning the patient, adjusting HOB, or mobilizing the patient out of bed to prevent severe overdrainage or underdrainage.
- **Waveform:** ICP value and waveform morphology are valid only when the EVD is clamped.
- **CSF Drainage:** CSF drainage is passive: occurs only when patient ICP exceeds EVD chamber height. Example: an EVD at 5 cmH<sub>2</sub>O drains more readily than one at 20 cmH<sub>2</sub>O.
- **Normal CSF Flow:** Normal CSF production is ~20 mL/hr (~500 mL/day). Drainage >20 mL/hr should trigger immediate assessment for overdrainage or chamber level escalation.
- **Weaning:** Gradual escalation of drainage setting by 5 cmH<sub>2</sub>O per day. After +20 cmH<sub>2</sub>O, EVD should be clamped & head CT obtained to evaluate ventricular caliber. Neurologic examination, CSF output, and ICP waveform should be assessed daily.

## COMPLICATIONS

- **Overdrainage (>20 mL/hr):** Risk of subdural hematomas (bridging vein tearing), ventricular collapse (slit ventricles), or upward cerebellar herniation.
- **Underdrainage:** Risk of worsening hydrocephalus, brain compression, or elevated ICP. Troubleshoot for system kinks, blood clots, air locks, or malpositioned stopcocks.